



VM0048 – Avoided Deforestation estimates using Google Earth Engine and the ClarkLabs UDef-ARP tool

A GUIDE FOR DEVELOPING EMISSIONS BASELINES FOR VM0048

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Introduction

The largest global voluntary carbon standard body, VERRA has been refining and consolidating their numerous avoided unplanned deforestation carbon accounting methodologies to a single methodology for accounting for avoided emissions from reducing unplanned deforestation and forest degradation rates. The full implementation of the new methodology (VM0048) has been faced by significant delays, however most information about the new methodology is now publicly available.

The REDD+ methodology contains some significant deviations from the historical REDD+ methodologies and approaches, but key differences include:

1. Carbon project developers no longer create their own deforestation rate estimations. VERRA are now responsible for developing and providing the activity data and risk maps for projects to use as a baseline against which to measure project impacts. This is an effort to attempt to ensure that there is consistency in rates total baseline deforestation, preventing project developers from over-inflating baseline rates of deforestation and allows better capture of leakage and the local displacement of deforestation within a landscape.
2. Future rates of deforestation within the project area are no longer calculated based on the simple projection of a 10-year historical rates of deforestation calculated in a project reference period for a 10-year period, but rather on a jurisdictional risk mapping and deforestation area allocation approach.

At time of writing, there are still several challenges involved in the development of Carbon projects that utilize this methodology, specifically:

1. Carbon credit yields become very uncertain beyond a 6-year forecast as the jurisdictional deforestation rates are valid for a 6-year basis.
2. Verra has hired consultants to develop the activity data and risks maps on a country-by-country basis, with the majority of jurisdictions data still to be released and has been beset by delays.
3. There are no guidelines on how JNR, VM0048 and Equitable Earths M.002 methodology can be reconciled, introducing uncertainty into the VCM.

This guide has been designed to assist analysts with developing *ex-ante* estimates of the climate impact associated with Avoided unplanned deforestation in accordance with the Verra VM0048 Methodology.

Under VM0048, future deforestation rates are allocated to the project region based on the outputs of a deforestation risk prediction model, which has been trained on the spatial patterns of forest vs non-forest of the entire jurisdictional area (i.e. the country or province that the project occupies) over a 10-year historical baseline (known as the baseline jurisdictional activity data). The predictive algorithm is then used to project forwards the risk that an area will be deforested in the next 6-year period. This process involves three stages:

1. Development of the Forest Cover Benchmark Map
2. Development of a Jurisdictional Deforestation Risk map
3. Allocation of the expected jurisdictional deforestation rates to the risk categories.

In the absence of the official VERRA supplied risk maps and allocated jurisdictional activity data, The TNC Africa Carbon Markets technical team have developed versions of the activity data and risk maps using

TNC developed a Google Earth Engine Risk Mapping toolkit (based off the guidelines in Verra VT0007), which is used for developing the jurisdictional Risk map. The outputs of the risk map are then used as inputs to the ClarkLabs Udef-ARP tool.

This approach allows analysts to develop a robust baseline in the absence of the published Verra Activity Data for the country of interest.

An important caveat is that the data used in these scripts are derived using globally available landcover datasets and as such may not align with your jurisdictions forest cover benchmark map definitions. If your country has better Landcover maps than the ones used in the script worked example, it is advisable to use these instead, and there is provision for them to replace the global data within the script.

Consequently, the accuracy of the modelled future deforestation rates in this example may deviate from the final model outputs published by Verra and the results of the script can only to be considered preliminary, with subsequent credit estimates to be performed when the official data becomes available.

Methodology

Benchmark vs Alternative Models

Due to the complexity involved in predicting future deforestation events, the recommendation of the VT0007 module is to test what is termed the benchmark model against a more complex and informed model, termed the alternative model. This then provides the end user with a clear understanding of the performance of their model in relation to a standard industry benchmark.

- Benchmark Model - a simple model which uses distance from forest edge to predict future deforestation events
- Alternative Model – A more complex model, often using a suite of input parameters (distance from forest edge, distance from rivers, distance from roads, distance from cropland, landscape travers ability, ecosystem type, Aboveground Biomass Density, etc.) to predict the future likelihood of deforestation.

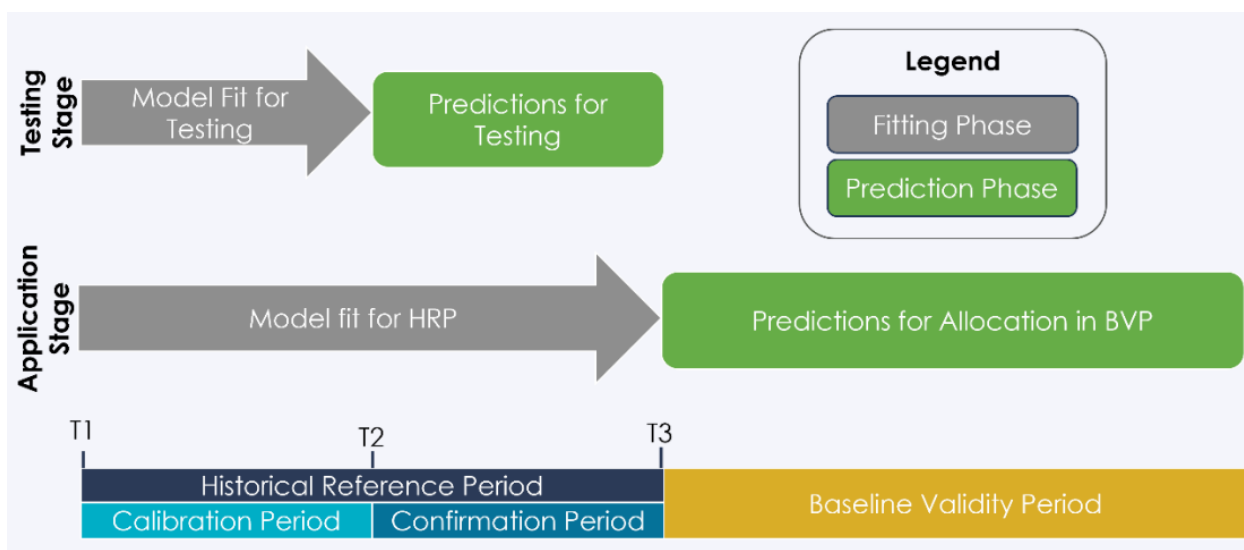
Temporal periods

The VT007 module of VM0048 requires that data used in the development of the deforestation baseline is trained and tested on data for 3 temporal periods:

T3, the baseline validity period (VP or BVP) – a 6-year period where the modelled baseline deforestation results are deemed valid.

T1, the start of the historical reference period and the start of the calibration period – a period 10 years prior to the start of the BVP.

T2, the start of the calibration period – a period close to the mid-point of the historical reference period



The analysis script that has been developed adds an additional temporal period 5 years prior to t1 (Called t0 in this document). The reason for this is that the deforestation probability algorithm requires distance from historical deforestation as an input and as such, this distance needs to be calculated.

Development of the Forest cover Benchmark Model

The forest cover Benchmark map is an important input that is used to define the forest cover for the different epochs of the analysis. It is a jurisdictional raster layer of forest cover, where pixel values are allocated according to the following categories:

HRP Start Year Forest	HRP Midpoint Forest	HRP End Year Forest	FCBM Index	Allocated Risk Mapping Role			Interpretation
				Test Risk Map	HRP Risk Map	Validity Risk Map	
No	No	No	1	NF	NF	NF	Stable Non-Forest
No	No	Yes	2	NF	NF	NF	Stable Non-Forest
No	Yes	No	3	NF	NF	NF	Stable Non-Forest
No	Yes	Yes	4	NF	NF	NF	Stable Non-Forest
Yes	Yes	Yes	5	F	F	F	Stable Forest
Yes	No	No	6	F	NF	NF	Deforested T1-T2
Yes	No	Yes	7	F	NF	NF	Deforested T1-T2
Yes	Yes	No	8	F	F	NF	Deforested T2-T3

These forest categories are then used to develop the inputs for the risk model as well as to test the risk model and deforestation allocation process against the standardized benchmark model.

Risk model inputs

The risk model utilizes the following datasets as inputs:

#	Input	Description
1	Deforestation Events	A raster layer showing the areas where deforestation occurred in forest areas between 2000 and each analytical period (t0, t1, t2, t3)
2	Distance From Deforestation events	A raster layer where each pixel represents the Euclidean distance from the nearest deforestation event
3	Forest Cover	A raster layer showing the areas where canopy cover was greater than 15% in 2000 (input – Hansen treecover) and landcover is a forest class (input - GLC_FCS30D – the Harmonised Global Landcover 1985-2022) and had not been flagged as deforestation (input Hansen lossyear) - for each analytical period (t0, t1, t2, t3).
4	Forest Edge Distance	A raster layer showing the Euclidean distance from the forest edge for each analytical period (t0, t1, t2, t3).
5	Cropland areas	A raster layer showing current cropland coverage input - GLC_FCS30D – the Harmonised Global Landcover 1985-2022) for each analytical period (t0, t1, t2, t3).
6	Cropland distance	A raster layer showing the Euclidean distance from the nearest cropland pixel for each analytical period (t0, t1, t2, t3).
7	Accessibility	A raster layer estimating the land-based travel time to the nearest densely-populated area (Malaria Atlas Project)
8	elevation	A raster layer showing the height above sea level (input – Shuttle Radar Topograph Mission)
9	slope	A raster layer derived from the SRTM layer showing the slope steepness in degrees
10	Aboveground woody biomass density	Pixel estimates of Aboveground woody biomass density for each analytical period (t0, t1, t2, t3) – Source ESA biomass mission data
11	Population	Pixel estimates of population density for each analytical period (t0, t1, t2, t3) – Source ESA biomass mission data
12	Population distance	A raster layer for each analytical period (t0, t1, t2, t3), showing the Euclidean distance from population areas where the population is greater than 1

Training and testing the Risk Model.

The risk model training process follows a typical machine learning pipeline where an array of labelled explanatory variables are split into training and validation sets, with the training set used to train a random forest classifier, which is then used to predict the likelihood of a pixel being deforested.

Deforestation is a multi-temporal phenomenon and as such, the labels need to be from a time step that is after the explanatory variables. Within the script, this is achieved using a 5-year time step, with all deforestation events occurring within the subsequent epoch used as labels, (i.e. t0 – t1 = explanatory variables, deforestation events between t1 – t2 = labels).

After the classifier has been trained, the model is then tested on the validation set (i.e. unseen data where we know what the label should be) and we compare the prediction to the actual class value.

All of these steps are performed automatically within the script, with little need for additional user inputs.

In some environments this approach has achieved an overall accuracy of 78% however jurisdictions may have more fragmented patterns of deforestation which are more challenging to predict. In these cases it may be necessary to add additional explanatory variables.

After running the script in google earth engine – it will export all the required files for the Deforestation area allocation to a specified google drive folder.

Deforestation area allocation

After the risk model has been trained, there are additional steps that need to be performed to assess the models performance and allocate deforestation area to the forest classes. The expected jurisdictional activity data (ha) is the sum of the forest area that a jurisdiction is expected to lose under the baseline scenarios during the Baseline validity period. The analysis script used to create the risk model inputs estimates this value based on a linear regression of historical annual jurisdictional deforestation rates (deforestation detected using hansen data), however if there are alternative deforestation estimates those should be used.

The Model testing and deforestation area allocation is performed using the ClarkLabs Udef-ARP toolkit (Available from <https://github.com/ClarkCGA/UDef-ARP/tree/main>)

This tool requires the following raster inputs:

Testing Stage

Fitting Phase (Calibration period) – this stage is used to develop and test the the negligible risk threshold of the benchmark model (i.e. the distance from forest edge where deforestation rates drop to near zero). Inputs include:

1. Distance from forest edge at the start of the calibration period (t1) in meters
2. Areas deforested between t1 and t3 (binary)
3. A jurisdictional mask (binary)
4. A mask of forest areas at the start of t1 (binary)
5. Risk of deforestation between t1 and t2 (values between 0 and 1)
6. Administrative divisions (integer)
7. Areas deforested between t1 and t2 (binary)

During the testing phase, the Calibration vulnerability map for the benchmark model is derived from distance to forest edge and a Negligible risk threshold that the tool calculates. The user is expected to run the benchmark model first and save the outputs to a specified working folder for the Benchmark model. this process is then repeated for the alternative model, which uses the Earth engine derived Risk maps as inputs.

The Fitting phase will then perform a fit assessment on the vulnerability map from the calibration period (needs to be repeated for both Benchmark and the Alternative model) which will determine how well the model predicted deforestation in the Calibration period.

The prediction phase (Confirmation period) – this stage is used to determine how well the alternative model compares to the benchmark model over the Confirmation period (i.e. $t_2 - t_3$). This is to test the risk model performance compared to the benchmark model over multiple time periods on unseen data.

Inputs include:

1. Distance from forest edge at the start of the confirmation period (t_2) in meters
2. The negligible risk threshold (Calculated during the testing phase)
3. Risk of deforestation between t_2 and t_3 (values between 0 and 1)
4. A jurisdictional mask (binary)
5. A mask of forest areas in t_2 (binary)
6. Administrative divisions (integer)
7. Areas deforested between t_2 and t_3 (binary)
8. Areas deforested between t_1 and t_2 (binary)
9. A mask of forest areas at the start of t_1 (binary)

Once the Testing stage is complete, there will be outputs which demonstrate model performance allowing for suitable adjustments to be made to the ex-ante carbon estimates based on selecting a conservative lower bound estimate of deforestation rates.

Application Stage

Fitting Phase (Historical Reference Period)

1. Distance from forest edge at the start of the Historical reference period (t_1) in meters
2. Risk of deforestation at t_1
3. A jurisdictional mask (binary)
4. A mask of forest areas at the start of t_1 (binary)
5. Administrative divisions (integer)
6. Areas deforested between t_1 and t_3 (binary)

Prediction Phase – The Deforestation Baseline

1. Distance from forest edge at the start of the Baseline validity period (t_3) in meters
2. Risk of deforestation at t_3
3. A jurisdictional mask (binary)
4. A mask of forest areas at the start of t_3 (binary)
5. Administrative divisions (integer)
6. Expected Jurisdictional deforestation area (ha)

The output of the Udef-ARP toolkit is a pixel prediction of annual deforestation area (ha) for the entire jurisdiction.

Baseline Emissions Scenario

To calculate the project level baseline emissions scenario, the per pixel prediction of annual deforestation area (ha) for the entire jurisdiction can be clipped to the project region. The resulting clipped raster can then be multiplied by an estimate of the aboveground woody biomass data, which can then be converted to CO₂ equivalents using an appropriate conversion factor. Zonal statistics can then be gathered by summing the pixels within the various project regions which will provide the final emissions baseline from which ex-ante estimates can be derived.

Tool Walkthrough

This walkthrough assumes that the user is proficient in the use of google earth engine as well as the installation and running of Anaconda Environments – If you are uncomfortable with either of these, then contact me via Hayden.Wilson@tnc.org and I will do my best to assist you with the process.

Google Earth Engine VM0048_JNR Riskmap.js script

Access the a snapshot of the Earth Engine script (July 2026 version) directly via <https://code.earthengine.google.com/d30bcb9c61ab4b50e66530319e88a285>

Latest version of the script is available via the TNC Africa Carbon Github page <https://github.com/HaydenTWilson/TNC-Africa-Carbon/tree/main/>

Configuration steps:

1. Select which jurisdiction you would like to model by removing the geometry import, called point and replacing it with your own geometry called point in your desired jurisdiction
2. Provide a name for the folder you would like to export the model outputs to (the default is 'eSwatini', but you should update that to the name of the country you are modelling)

Save the script and push the 'Run' button on the command bar – the model should now undergo the training and testing phase and set the export tasks for the various inputs for the UDef-ARP toolkit.

Troubleshooting and adjusting parameters:

There are occasions where the jurisdiction may be too big to model at once – in which case the model region should be split into smaller pieces to model effectively.

Export and Mosaicing

After Earth Engine has exported the Geotiffs they should be ready to download and use in the UDef-ARP toolkit without modification. If the Jurisdiction being modelled is large, Earth Engine may split the Geotiffs into smaller tiles. These will need to be mosaiced before use in the UDef-ARP tool. It is very important to ensure that processed Geotiffs do not contain null values and that the raster extent, pixel size and co-ordinate system are identical.

ClarkLabs Udef-ARP toolkit

(Available from <https://github.com/ClarkCGA/UDef-ARP/tree/main>)

The Udef-ARP Tool is run via a GUI. The outputs of the Google Earth Engine script have been configured to match the input requirements of the UDef-ARP tool.

Table 1: Inputs for UDef-ARP Matched to the files exported from Earth engine

#	Stage	Phase	Process	Model	UDef-ARP input	Earth Engine Export File	Comment
1	Testing	Fitting Phase (CAL)	Vulnerability Mapping	Benchmark	Map of Distance From Forest Edge in the CAL	CAL_Dist.tif	
2	Testing	Fitting Phase (CAL)	Vulnerability Mapping	Benchmark	Map of Deforestation in the HRP	HRP_Deform.tif	
3	Testing	Fitting Phase (CAL)	Vulnerability Mapping	Benchmark	Mask of the non-excluded Jurisdiction	Jurisdiction.tif	
4	Testing	Fitting Phase (CAL)	Vulnerability Mapping	Benchmark	Negligible Risk Threshold (m)	N/A	Calculated in the Tool
5	Testing	Fitting Phase (CAL)	Vulnerability Mapping	Alternative Model	Empirical Transition potential [0,1] for the CAL	CAL_Riskmap.tif	
6	Testing	Fitting Phase (CAL)	Vulnerability Mapping	Alternative Model	Mask of the Forest Areas in CAL	CAL_Forest.tif	
7	Testing	Fitting Phase (CAL)	Allocated Risk Mapping	Both Models	Map of Administrative Divisions	Admin.tif	
8	Testing	Fitting Phase (CAL)	Allocated Risk Mapping	Both Models	Vulnerability map for the CAL	N/A	Calculated in Tool during Vulnerability Mapping
9	Testing	Fitting Phase (CAL)	Allocated Risk Mapping	Both Models	Map of Deforestation in the CAL	CAL_Deform.tif	
10	Testing	Fitting Phase (CAL)	Model Fit Assessment	Both Models	Fitted Density Map for the CAL	N/A	Calculated in Tool during Allocated Risk Mapping
11	Testing	Fitting Phase (CAL)	Model Fit Assessment	Both Models	Mask of the non-excluded Jurisdiction	Jurisdiction.tif	

12	Testing	Fitting Phase (CAL)	Model Fit Assessment	Both Models	Map of Deforestation in the CAL	CAL_Deform.tif	
13	Testing	Prediction Phase (CNF)	Vulnerability Mapping	Benchmark	Map of Distance From Forest Edge in the CNF	CNF_Dist.tif	
14	Testing	Prediction Phase (CNF)	Vulnerability Mapping	Benchmark	Negligible Risk Threshold in the HRP	N/A	Calculated during testing fitting phase of the Benchmark Model
15	Testing	Prediction Phase (CNF)	Vulnerability Mapping	Alternative	Empirical Transition potential [0,1] for the CNF	CNF_Riskmap.tif	
16	Testing	Prediction Phase (CNF)	Vulnerability Mapping	Alternative	Mask of the non-excluded Jurisdiction	Jurisdiction.tif	
17	Testing	Prediction Phase (CNF)	Vulnerability Mapping	Alternative	Mask of the Forest Areas in CAL	CAL_Forest.tif	
18	Testing	Prediction Phase (CNF)	Allocated Risk Mapping	Both Models	Map of Administrative Divisions	Admin.tif	
19	Testing	Prediction Phase (CNF)	Allocated Risk Mapping	Both Models	CAL Relative Frequency Table	N/A	Calculated in tool during the Fitting Phase
20	Testing	Prediction Phase (CNF)	Allocated Risk Mapping	Both Models	Vulnerability map for the CNF	N/A	Calculated in Tool during Vulnerability Mapping
21	Testing	Prediction Phase (CNF)	Allocated Risk Mapping	Both Models	Map of Deforestation in the CNF	CNF_Deform.tif	
22	Testing	Prediction Phase (CNF)	Model Prediction Assessment	Both Models	Adjusted Predicted CNF Density Map	N/A	Output from Allocated Risk Mapping
23	Testing	Prediction Phase (CNF)	Model Prediction Assessment	Both Models	Mask of the non-excluded Jurisdiction	Jurisdiction.tif	
24	Testing	Prediction Phase (CNF)	Model Prediction Assessment	Both Models	Map of Deforestation in the CNF	CNF_Deform.tif	

25	Testing	Prediction Phase (CNF)	Model Prediction Assessment	Both Models	Map of Deforestation in the CAL	CAL_Deform.tif	
26	Testing	Prediction Phase (CNF)	Model Prediction Assessment	Both Models	Mask of the Forest Areas in CAL	CAL_Forest.tif	
27	Application	Fitting Phase (HRP)	Vulnerability Mapping	Benchmark	Map of Distance From Forest Edge in the HRP	CAL_Dist.tif	
28	Application	Fitting Phase (HRP)	Vulnerability Mapping	Benchmark	Negligible Risk Threshold (m)	N/A	Calculated in the Tool
29	Application	Fitting Phase (HRP)	Vulnerability Mapping	Alternative	Empirical Transition potential [0,1] for the HRP	CAL_Riskmap.tif	
30	Application	Fitting Phase (HRP)	Vulnerability Mapping	Alternative	Mask of the non-excluded Jurisdiction	Jurisdiction.tif	
31	Application	Fitting Phase (HRP)	Vulnerability Mapping	Alternative	Mask of the Forest Areas in HRP	CAL_Forest.tif	
32	Application	Fitting Phase (HRP)	Allocated Risk Mapping	Both Models	Map of Administrative Divisions	Admin.tif	
33	Application	Fitting Phase (HRP)	Allocated Risk Mapping	Both Models	Vulnerability map for the HRP	N/A	Calculated in Tool during Vulnerability Mapping
34	Application	Fitting Phase (HRP)	Allocated Risk Mapping	Both Models	Map of Deforestation in the HRP	HRP_Deform.tif	
35	Application	Prediction Phase (VP)	Vulnerability Mapping	Benchmark	Map of Distance From Forest Edge in the VP	BVP_Dist.tif	
36	Application	Prediction Phase (VP)	Vulnerability Mapping	Benchmark	Negligible Risk Threshold in the HRP (m)	N/A	Calculated in the Tool
37	Application	Prediction Phase (VP)	Vulnerability Mapping	Alternative	Empirical Transition potential [0,1] for the VP	BVP_Riskmap.tif	

38	Application	Prediction Phase (VP)	Vulnerability Mapping	Alternative	Mask of the non-excluded Jurisdiction	Jurisdiction.tif	
39	Application	Prediction Phase (VP)	Vulnerability Mapping	Alternative	Mask of Forest Areas in the VP	BVP_Forest.tif	
40	Application	Prediction Phase (VP)	Allocated Risk Mapping	Both Models	Map of Administrative Divisions	Admin.tif	
41	Application	Prediction Phase (VP)	Allocated Risk Mapping	Both Models	HRP Relative Frequency Table	N/A	Calculated in tool during the Fitting Phase
42	Application	Prediction Phase (VP)	Allocated Risk Mapping	Both Models	Vulnerability map for the VP	N/A	Calculated in Tool during Vulnerability Mapping
43	Application	Prediction Phase (VP)	Allocated Risk Mapping	Both Models	Expected Jurisdictional Activity	BVP_expected_annual_deforestation.csv	Use the value from Annual_deforestation_BVP column